Optional projects

March 2

The following is a list of projects which you may turn in to replace some number of quiz scores. The exact number depends on the project. Please note, I will only accept **three** solutions for each project (call them "bounties"). Solutions must be written in complete sentences and all details must be provided. As soon as I receive three solutions (complete) I will remove the project from this list.

Project 1: *Lagrange polynomials.* Replaces one quiz score, 3 bounties remain.

Let $c_0, c_1, \ldots, c_n$ be distinct real numbers. Define, for $i = 0, \ldots, n$ the functions $f\colon \mathbb{R} \to \mathbb{R}$ by the formula

$$f_i(x) = \prod_{k \neq i} \frac{x - c_k}{c_i - c_k}. \tag{1}$$

(a) Show for each i that $f_i(x) \in \mathsf{P}_n$.

(b) Show that $\{f_0, \ldots, f_n\} \subset \mathsf{P}_n$ is linearly independent and hence a basis.

(c) Suppose that $g \in \mathsf{P}_n$ is any polynomial such that $g(c_0) = g(c_1) = \cdots = g(c_n) = 0$. Show that g is the zero polynomial.

Project 2: *The dimension of* $\mathbf{Hom(V, W)}$. Replaces one quiz score, 3 bounties remain.

In class, we have seen that the set $\mathrm{Hom}(\mathsf{V}, \mathsf{W})$ of all linear transformations from V to W form a vector space with the obvious rules of addition and scalar multiplication. In this problem, you will compute $\dim \mathrm{Hom}(\mathsf{V}, \mathsf{W})$.

(a) Set $\dim \mathsf{V} = n, \dim \mathsf{W} = m$. Let $\beta = \{v_i\}$ be a basis for V and $\{w_j\}$ be a basis for W. For any $i = 1, \ldots, n$ and $j = 1, \ldots, m$ define $\varphi_{i,j}\colon \beta \to \mathsf{W}$ by the formula:

$$\varphi_{i,j}(v_k) = \begin{cases} w_j, & i = k \\ 0, & \text{else} \end{cases} \tag{2}$$

Argue why $\varphi_{i,j}$ determines a linear transformation $\mathsf{V} \to \mathsf{W}$.

(b) Show that the collection $\{\varphi\}_{i=1,\ldots,n;j=1,\ldots,m} \subset \mathrm{Hom}(\mathsf{V}, \mathsf{W})$ is linearly independent.

(c) Show that the collection from part (b) is a basis. Conclude that

$$\dim \mathrm{Hom}(\mathsf{V}, \mathsf{W}) = (\dim \mathsf{V}) \cdot (\dim \mathsf{W}). \tag{3}$$

Project 3: *Multilinear functions.* Replaces one quiz score, 3 bounties remain.

If A, B are any sets, we can form a new set called the product set denoted $A \times B$. It's defined by

$$A \times B = \{(a, b) \mid a \in A, b \in B\}. \tag{4}$$

(a) If $\#A = n$ and $\#B = m$ then $\#(A \times B) =?$.

(b) Now, consider three sets A, B and C. And let $f\colon A \times B \to C$ be any function. If $a \in A$ is a fixed element, define $f_{a,}\colon B \to C$ by the formula $f_{a,}(b) = f(a, b)$. And, if $b \in B$ is a fixed element, define $f_{,b}\colon A \to C$ by the formula $f_{,b}(a) = f(a, b)$. Show that $f\colon A \times B \to C$ is surjective if there exists some $a \in A$ such that $f_{a,}\colon B \to C$ is surjective.

(b) Consider three vector spaces V, W, and U. Define a *bilinear* transformation to be a function

$$\mathsf{T}\colon \mathsf{V} \times \mathsf{W} \to \mathsf{U} \tag{5}$$

such that the two conditions hold:

(i) $\mathsf{T}_{\mathbf{0}_V,}\colon \mathsf{W} \to \mathsf{U}$ is a linear transformation, where $\mathbf{0}_V \in \mathsf{V}$ is the zero vector in V.

(ii) $\mathsf{T}_{,\mathbf{0}_W}\colon \mathsf{V} \to \mathsf{U}$ is a linear transformation, where $\mathbf{0}_W \in \mathsf{W}$ is the zero vector in W.

Show that the function

$$\mathsf{T}\colon \mathbb{R}^2 \times \mathbb{R}^3 \to \mathbb{R} \tag{6}$$

defined by

$$\mathsf{T}\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) = x_1 y_2 - x_2 y_1 + 3x_1 y_3$$

is bilinear.

(c) Generalize the definition above to define the notion of a *k-multilinear transformation* $\mathsf{T}\colon \mathsf{V}_1 \times \cdots \times \mathsf{V}_k \to \mathsf{U}$. Provide one nontrivial example in the case $k = 3$.

Project 3: *Determinant in dimension two.* Replaces two quiz scores, 3 bounties remain.

This problem uses the concept of a bilinear map defined in the previous project. Let V be a two-dimensional vector space. A *two-form* on V is a bilinear map

$$\omega\colon \mathsf{V} \times \mathsf{V} \to \mathbb{R} \tag{7}$$

with the property that $\omega(v, w) = -\omega(w, v)$ for all $v, w \in \mathsf{V}$.

(a) Show that there exists a nontrivial two-form on V (Hint: pick a basis).

(b) Let ω_0 be the nontrivial two-form you found in part (a). Show that any other two-form ω satisfies $\omega(v, w) = \delta\omega_0(v, w)$ for all $v, w \in \mathsf{V}$ where δ is some scalar (the scalar is independent of v, w).

(c) Suppose that $\mathsf{T}\colon \mathsf{V} \to \mathsf{V}$ is a linear transformation and let ω be any two-form on V. Define the function

$$\omega_{\mathsf{T}}\colon \mathsf{V} \times \mathsf{V} \to \mathbb{R} \tag{8}$$

by the formula $\omega_{\mathsf{T}}(v, w) = \omega(\mathsf{T}(v), \mathsf{T}(w))$. Show that ω_T is a two-form.

(d) By part (c), there exists a δ_{T} such that

$$\omega_{\mathsf{T}}(v, w) = \delta_{\mathsf{T}}\omega(v, w). \tag{9}$$

Show that δ_{T} is well-defined; that is, it does not depend on the two-form ω and so only depends on T.

(e) Consider the example of $\mathsf{V} = \mathbb{R}^2$ and the linear transformation

$$\mathsf{T}\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix} \tag{10}$$

for some scalars a, b, c, d. Compute δ_{T} in this example. Does the expression look familiar?